

FILTER DESIGN TEST CASE

Coupled Resonator Filter - Band 20 (800 MHz) for Automotive V2X Communication
Design Validation and Performance Characterization

DOCUMENT INFORMATION

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1. OBJECTIVE

This document describes the design, simulation, and characterization of a Coupled Resonator Filter filter targeting Band 20 (800 MHz) for Automotive V2X Communication. The filter is designed on PZT on Si substrate with a target center frequency of 884.6 MHz and bandwidth of 148.0 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the Automotive V2X Communication module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: Coupled Resonator Filter
- Frequency Band: Band 20 (800 MHz)
- Application: Automotive V2X Communication
- Substrate: PZT on Si
- Package: SiP (System in Package)
- Design Tool: Matlab RF Toolbox R2024b
- Design Center: San Jose Advanced Design Lab

This specification applies to all variants of the Coupled Resonator Filter design targeting Band 20 (800 MHz). Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency:	884.6 MHz
Bandwidth (3 dB):	148.0 MHz
Insertion Loss Target:	≤ 3.6 dB
Return Loss Target:	≥ 20.0 dB
Out-of-Band Rejection:	≥ 50.0 dB
Isolation (Duplexer):	≥ 37.0 dB
Q Factor:	2341
Temperature Coefficient:	-24.3 ppm/°C
Die Size:	2.7 x 1.3 mm
Wafer Size:	8-inch
Number of Resonators:	9
Electrode Material:	W

Passivation: Si3N4

4. TEST PROCEDURE

1. Open Matlab RF Toolbox R2024b and load the Coupled Resonator Filter template project.
2. Define the substrate stack: PZT on Si with specified layer thicknesses.
3. Set design targets: center freq 884.6 MHz, BW 148.0 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 3.6 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (SiP (System in Package)).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (8-inch, PZT on Si).
10. Perform on-wafer probing using Rohde & Schwarz FSW Signal Analyzer.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 2654 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 3.6 dB
- Return loss in passband shall be >= 20.0 dB
- Out-of-band rejection at +/- 222 MHz offset >= 50.0 dB
- Group delay variation across passband shall be <= 4 ns
- Temperature drift over -40°C to +85°C shall be <= 2.7 MHz
- Power handling shall be >= +34 dBm at 1 dB compression
- ESD tolerance >= 2000 V (HBM)
- Die yield on qualification lot >= 98%

6. TEST RESULTS

Measurement Date: 2024-06-16
Devices Tested: 91
Insertion Loss (meas): 3.33 dB
Return Loss (meas): 22.4 dB
Rejection (meas): 52.5 dB
Isolation (meas): 41.7 dB
Q Factor (meas): 2341
Group Delay Variation: 5.0 ns
Power Handling: +26 dBm
Die Yield: 94.3%

Result Classification: FAIL

7. OBSERVATIONS AND FINDINGS

- a) The Coupled Resonator Filter design demonstrated below-target performance for Band 20 (800 MHz).
- b) Insertion loss is marginally above target; electrode thickness optimization recommended.
- c) Out-of-band rejection exceeded the minimum requirement.
- d) Temperature stability was within specification across full range.
- e) No delamination observed after 1000-cycle thermal shock test.

8. CORRECTIVE ACTIONS

- 1. Optimize resonator geometry to reduce insertion loss by ~0.3 dB.
- 2. Re-run EM simulation with refined mesh for better accuracy.
- 3. Escalate to advanced materials team for alternative piezoelectric stack.

9. SIGN-OFF

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